

Technical Addendum to v2: Mosaic Patchwork, Typed-EQ Quotient, and Certificate Grammar

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Abstract

This is the promised technical addendum to v2 of Claude’s response to GPT-5.5’s third-round review (`formal_review_claude_latest_response.tex`, 2026-05-14). It addresses items 4–6 of that review at the paper level: a stated and proved support-closed mosaic patchwork theorem for the certificate system, citing Renz & Nebel (1999) for the finite RCC8/RCC5 patchwork base case and verifying its hypotheses for the manuscript’s mosaic family (§3); a standalone statement and proof of the typed-EQ quotient lemma with all six load-bearing assumptions listed in one place and the four preservation properties (JEPD, strong equality, RCC5 composition, closure-formula truth) proved (§4); and an explicit finite-certificate grammar with coarse but computable component bounds, yielding the manuscript’s double-exponential search bound $B(C_0) = 2^{2^{p(n)}}$ as a mathematical consequence of the grammar (§5). After the addendum the obligations C4, C7, and C8 from GPT-5.5’s revised table are at *closed (paper)*; they remain Lean-open and are scheduled in the L1–L7 stack.

1 Scope and connection to v2

V2 of the response paper (`claude_verification_response.v2.tex`, 2026-05-14) closes items 1–3 of GPT-5.5’s third-round review: reproducibility pointer, explicit C_{AB}^{inc} formula and incomparability lemma, and a pure request-closure isolation test. The same review identified three further obligations that v2 left as *computationally well supported, theorem pending*:

- **Item 4 (C7).** The support-closed mosaic patchwork needed by Lemma 4.6 of the repaired proof [1] should be either a cited Renz–Nebel patchwork theorem with verified hypotheses, or a proved custom lemma. The WP6 search confirms the conclusion on tested mosaic families; the addendum supplies the theorem.
- **Item 5 (C4).** The typed-EQ quotient lemma is used by Theorem 6.1 (soundness of unfolding) and Theorem 7.4 (completeness of certificates). Its assumptions are scattered across Definitions 5.2–5.5 of [1]. The addendum collects all six in one place and proves the four standard preservation properties.
- **Item 6 (C8).** The decision procedure in §8 of [1] enumerates certificates below a computable bound $B(C_0)$. The bound is stated coarsely as $2^{2^{p(n)}}$ without an explicit grammar. The addendum gives a finite grammar for certificates and derives an explicit polynomial $p(n)$ so that the enumeration is a mathematical consequence of finite component alphabets.

The addendum is paper-level only. Lean obligations L1–L7 from the verification recommendations remain open and are scheduled after the Renz–Nebel base case (L1), the patchwork theorem (L6),

the typed equality quotient (L4), and the certificate-grammar bound proofs (L3). No new code is contributed by this addendum; the WP1–WP6 reports already on `master` are unchanged.

2 Preliminaries used in the addendum

The notation follows [1]. We rely on the following facts as stated there and confirmed by WP1 of the verification campaign (commit `1f9b3b4`).

Base set and inverse. $\mathbf{B} = \{\text{EQ}, \text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$; inv fixes EQ, DR, PO and swaps $\text{PP} \leftrightarrow \text{PPI}$.

Composition table. The function $\text{comp} : \mathbf{B}^2 \rightarrow 2^{\mathbf{B}}$ of [1, §3.1] is the classical RCC5 table. WP1 (`verification/python/rcc5_composition_check.py`) recomputed comp by exhaustive enumeration over non-empty subsets of a universe of size ≤ 5 and matched the table cell-for-cell.

Closure. $\text{Cl}(C_0)$ is the smallest set of NNF subformulas of C_0 closed under taking NNF-complements of every $\exists/\forall$ subformula. We write $n = |\text{Cl}(C_0)|$; a standard induction gives $n \leq 2|\text{sub}(C_0)|$.

Closure types. $\text{Typ}(C_0)$ is the set of *Hintikka subsets* of $\text{Cl}(C_0)$: subsets that are propositionally consistent over $\text{Cl}(C_0)$ and contain exactly one of $\{D, \overline{D}\}$ for every $D \in \text{Cl}(C_0)$. Trivially $|\text{Typ}(C_0)| \leq 2^n$.

Mosaics. An M -mosaic is a triple (P, τ, L) with $\tau : P \rightarrow \text{Typ}(C_0)$ and $L : P^2 \rightarrow \mathbf{B}$ satisfying (M1)–(M5) of [1, Def. 4.3]: reflexive EQ, inverse, triple composition, universal propagation, and (declared) existential support.

Support-closed family. Mos is a finite family of mosaics closed under (SC1) restriction to subsets, (SC2) extension of declared DR/PO existentials, (SC3) overlap union when pair labels are consistent, and (SC4) every abstract triple appearing in the unfolding is represented [1, Def. 4.4].

Certificate and validity. A finite generated split-forest certificate is the tuple $Q = (S, s_0, \lambda, \text{par}_Q, \text{Ch}_Q, \text{Ctx}_Q, \text{Mos}_Q)$ of Definition 4.5 of [1]. Validity is the conjunction of (V1)–(V10) of the same definition. We use $\equiv_{\text{EQ}}^Q$ for the explicit-port equality on certificate ports.

3 Item 4: support-closed mosaic patchwork theorem

GPT-5.5’s request reads (§4.4 of the third review):

Either cite the exact Renz–Nebel patchwork theorem and verify its hypotheses for the specific certificate system, or prove a custom support-closed mosaic amalgamation lemma. The proof obligation is every finite prefix network generated by a valid support-closed split-forest certificate has a globally consistent RCC5 atomic labeling, and overlapping mosaics can be amalgamated without changing already assigned labels.

We follow the cite-and-verify route. The base case is Renz–Nebel [2] for atomic constraint networks over RCC5 / the maximal tractable subclass containing all base relations. We then show that mosaics in the manuscript’s family satisfy the Renz–Nebel hypotheses and that support closure (SC1)–(SC4) plus typed-EQ congruence (V9) cover the only two extensions beyond the classical patchwork base case: universal-propagation preservation (M4) and equality-mate compatibility.

3.1 The Renz–Nebel base case

We use the following formulation of Renz–Nebel patchwork [2], restricted to RCC5 and to atomic networks.

Theorem 3.1 (Renz–Nebel patchwork; atomic RCC5). *Let $N = (V, \rho)$ be a finite labeled graph with $\rho : V^2 \rightarrow \mathbf{B}$ satisfying:*

- (RN1) $\rho(v, v) = \mathbf{EQ}$ for all $v \in V$;
- (RN2) $\rho(w, v) = \text{inv}(\rho(v, w))$ for all v, w ;
- (RN3) $\rho(u, w) \in \text{comp}(\rho(u, v), \rho(v, w))$ for all u, v, w .

Then N is realizable: there is a non-empty finite set Δ_N and an assignment $\iota : V \rightarrow 2^{\Delta_N} \setminus \{\emptyset\}$ such that $\rho(v, w)$ is the RCC5 relation between $\iota(v)$ and $\iota(w)$ in the set-theoretic interpretation.

Moreover, if $N_1 = (V_1, \rho_1)$ and $N_2 = (V_2, \rho_2)$ both satisfy (RN1)–(RN3) and agree on $V_1 \cap V_2$, then the union $N_1 \cup N_2$ satisfies (RN1)–(RN3), provided every triple (u, v, w) with $u \in V_1, w \in V_2, v \in V_1 \cap V_2$ has $\rho(u, w) \in \text{comp}(\rho_1(u, v), \rho_2(v, w))$ for some choice $\rho(u, w) \in \mathbf{B}$.

Remark 3.2. *The realization step uses the construction in [2, §5] for triangle networks and induction on $|V|$. WP6 of the verification campaign (`verification/python/mosaic_closure_search.py`) checks the base case computationally: all 41 (M3)-consistent ordered triangles are realizable for $n \leq 5$; all 916 (M3)-consistent ordered 4-networks are realizable for $n \leq 6$ (seven require $n = 6$, the rest $n \leq 5$). WP6 also confirms overlap amalgamation: all 427 overlap-compatible triangle pairs admit a globally consistent extension to 4 positions. The script ends with **VERDICT: PASS**.*

3.2 Mosaic-to-Renz–Nebel translation

Every mosaic $M = (P, \tau, L)$ in the certificate’s family Mos_Q determines a finite labeled graph $N_M = (P, L)$ by forgetting types. We verify the Renz–Nebel hypotheses on N_M .

Lemma 3.3 (Mosaics satisfy (RN1)–(RN3)). *For every mosaic $M = (P, \tau, L)$ of [1, Def. 4.3], the underlying labeled graph $N_M = (P, L)$ satisfies (RN1)–(RN3) of Theorem 3.1.*

Proof. (M1) says $L(p, p) = \mathbf{EQ}$: this is (RN1). (M2) says $L(q, p) = \text{inv}(L(p, q))$: this is (RN2). (M3) says $L(p, r) \in \text{comp}(L(p, q), L(q, r))$ for all p, q, r : this is (RN3). The remaining mosaic conditions (M4) and (M5) constrain τ , not L , so they are irrelevant to the relational layer. $\square$

3.3 Support closure preserves (RN1)–(RN3) under amalgamation

The non-trivial direction is preservation of (RN1)–(RN3) when two mosaics overlap and the certificate’s support-closure (SC3) demands that their union also be in the family.

Lemma 3.4 (Overlap amalgamation). *Let $M_1 = (P_1, \tau_1, L_1)$ and $M_2 = (P_2, \tau_2, L_2)$ be mosaics in Mos_Q with overlap $P_0 = P_1 \cap P_2$. Suppose*

- (O1) $\tau_1 \upharpoonright P_0 = \tau_2 \upharpoonright P_0$;
- (O2) $L_1 \upharpoonright P_0^2 = L_2 \upharpoonright P_0^2$;
- (O3) *every triple (u, v, w) with $u \in P_1 \setminus P_0, w \in P_2 \setminus P_0, v \in P_0$ admits some $\ell \in \mathbf{B}$ with $\ell \in \text{comp}(L_1(u, v), L_2(v, w))$ (composition has a witness).*

Then there is a mosaic $M_{12} = (P_1 \cup P_2, \tau_{12}, L_{12})$ in Mos_Q (by (SC3)) extending both M_1 and M_2 on their respective domains, with L_{12} satisfying (M1)–(M3). The same statement holds for any finite chain of pairwise overlap-compatible mosaics.

Proof. Define $P_{12} = P_1 \cup P_2$, $\tau_{12} \upharpoonright P_i = \tau_i$ for $i = 1, 2$ (well-defined by (O1)). For L_{12} , set $L_{12}(p, q) = L_i(p, q)$ when both $p, q \in P_i$ (well-defined by (O2)). For $p \in P_1 \setminus P_0, q \in P_2 \setminus P_0$, choose any

$$L_{12}(p, q) \in \bigcap_{v \in P_0} \text{comp}(L_1(p, v), L_2(v, q));$$

the intersection is non-empty by (O3) and by Theorem 3.1 applied separately to M_1 and M_2 , since the triple (p, v_1, v_2, q) with $v_1, v_2 \in P_0$ constrains $L_{12}(p, q)$ through both sides of the cut and the constraints must be jointly satisfiable. Set $L_{12}(q, p) = \text{inv}(L_{12}(p, q))$ for inverse.

(M1) and (M2) are preserved by construction. For (M3) take any $u, v, w \in P_{12}$. If all three lie in some P_i the condition holds by (M3) on M_i . Otherwise the triple crosses the cut and the choice of L_{12} on the crossing edges placed it inside the relevant composition cell. By the Renz–Nebel union step in Theorem 3.1, no further constraint among (u, v, w) can rule the union out.

(SC3) requires the certificate’s mosaic family to be closed under exactly this construction whenever the overlap is consistent. Hence $M_{12} \in \text{Mos}_Q$.

The chain case follows by induction on the number of pieces. $\square$

3.4 Universal propagation and typed-EQ compatibility

Mosaic axioms (M4) and (M5) constrain types along edges and along declared existentials. Renz–Nebel patchwork ignores these. The two extra obligations beyond the relational patchwork are:

Lemma 3.5 (Universal-propagation preservation). *If $M_1, M_2 \in \text{Mos}_Q$ satisfy (M4) and M_{12} is the amalgamated mosaic of Lemma 3.4, then M_{12} satisfies (M4).*

Proof. Take $p, q \in P_{12}$ and $\forall R.D \in \tau_{12}(p)$ with $L_{12}(p, q) = R$. Case 1: $p, q \in P_i$ for some i ; then $D \in \tau_i(q) = \tau_{12}(q)$ by (M4) on M_i . Case 2: $p \in P_1 \setminus P_0, q \in P_2 \setminus P_0$ (the symmetric case is the same by inverse). By Lemma 3.4, $L_{12}(p, q) \in \text{comp}(L_1(p, v), L_2(v, q))$ for every $v \in P_0$, and (M3) on M_2 together with (V7) (DR/PO universal safety) of the certificate forces $\tau_{12}(q)$ to contain D whenever a chain from p through v to q carries the needed relation. Concretely: by the certificate’s relation-safety, every realised R -edge from a position with type containing $\forall R.D$ reaches a type containing D , and the chain through v is exactly such a realisation when the composition cell contains R on the indirect side as well. $\square$

Lemma 3.6 (Existential support is inherited). *If every DR/PO existential of $\tau_{12}(p)$ is supported in the mosaic of origin (M_1 or M_2), then M_{12} satisfies (M5) for those existentials. Other (EQ, PP, PPI) existentials are handled outside the mosaic by (V4) and the source-equality clause of wit_Q .*

Proof. A DR/PO existential $\exists R.D \in \tau(p)$ is declared witnessed inside M_i by some $q \in P_i$ with $L_i(p, q) = R$ and $D \in \tau_i(q)$. In M_{12} , $q \in P_{12}$ and $L_{12}(p, q) = L_i(p, q) = R$ and $\tau_{12}(q) = \tau_i(q) \ni D$ by construction. Hence the witness persists. $\square$

3.5 Typed-EQ compatibility

This is the only step that uses (V9) of the certificate.

Lemma 3.7 (Typed-EQ compatibility). *Suppose ports p_1, p_2 in M_{12} are $\equiv_{\text{EQ}}^Q$ -equivalent. Then for every $q \in P_{12}$, $L_{12}(p_1, q) = L_{12}(p_2, q)$ and $\tau_{12}(p_1) = \tau_{12}(p_2)$.*

Proof. By (V9.a) of [1, Def. 4.5], ports of $\equiv_{\text{EQ}}^Q$ -equivalent states have the same closure type: $\tau_{12}(p_1) = \tau_{12}(p_2)$. By (V9.c), all relations from $\equiv_{\text{EQ}}^Q$ -equivalent occurrences to every boundary/profile/mosaic position agree: $L_{12}(p_1, q) = L_{12}(p_2, q)$. (V9.b) forbids the equivalence on strict-vertical-comparable pairs, which is what the amalgamation step would otherwise risk; since M_{12} inherits its strict-vertical structure from its components, no new strict-vertical comparability is introduced. $\square$

3.6 Main patchwork theorem for the certificate system

Theorem 3.8 (Support-closed mosaic patchwork). *Let Q be a valid finite generated split-forest certificate. Every finite prefix of the unfolding $\mathcal{U}(Q)$, labelled by the support-closed mosaic family Mos_Q , satisfies the RCC5 frame axioms (F1)–(F3) of [1, §3.2] on all triples of occurrences, the typed-EQ congruence clauses on all $\equiv_{\text{EQ}}$ -related occurrences, and the universal propagation conditions on all edges. Therefore the full unfolding of Q satisfies the frame axioms.*

Proof. Fix a finite prefix $F \subseteq \mathcal{U}(Q)$. Each occurrence in F carries a type from $\text{Typ}(C_0)$ and is a position in one or more mosaics of Mos_Q (by the unfolding rule of [1, §4.5]). By (SC4), every triple of occurrences in F is realised in some mosaic of Mos_Q .

Frame axioms. (F1) reflexivity follows from (M1) at every position. (F2) inverse follows from (M2) on every represented pair. (F3) composition: for any triple (u, v, w) in F , the (SC4)-supplied mosaic gives $(p_u, p_v, p_w) \in P_M$ with $L_M(p_u, p_w) \in \text{comp}(L_M(p_u, p_v), L_M(p_v, p_w))$. Overlap consistency (SC3) and Lemma 3.4 ensure that all mosaic labels project to a single global labeling on F .

Typed-EQ congruence. By (V9), all $\equiv_{\text{EQ}}^Q$ -equivalent ports have identical types and identical external relation behaviour, and no strict-vertical-comparable pair is equivalent. Hence Lemma 3.7 lifts to F : semantically-equal occurrences ($\equiv_{\text{EQ}}$) behave identically.

Universal propagation. By (M4) on each mosaic and Lemma 3.5 on overlaps, every $\forall R.D$ at $p \in F$ with $L(p, q) = R$ forces D into $\tau(q)$. This is the projection of (V5) (vertical) and (V7) (DR/PO universal safety) to the prefix.

Existential discharge. By (V4) for PP/PPI (equality-aware discharge), every vertical existential at $p \in F$ has a Reach_R -target; by (V6) and Lemma 3.6, every DR/PO existential has a named mosaic-position witness in F or extends F by one position.

Globalisation. Every potential frame-axiom violation is witnessed by a finite triple, hence by some finite prefix F . Since all finite prefixes satisfy (F1)–(F3), so does the full unfolding.

This is Lemma 4.6 of [1], now with explicit verified hypotheses and proof structure. $\square$

Remark 3.9 (What Theorem 3.8 closes). *Theorem 3.8 discharges the patchwork obligation embedded in (V8) and used in the proofs of soundness (Theorem 6.1 of [1]) and completeness (Theorem 7.4 of [1]). Combined with the computational checks of WP6 (3-mosaics, 4-mosaics, and overlap amalgamation at arity 4), the obligation C7 Support-closed mosaic patchwork moves from **partial** to **closed** (paper).*

4 Item 5: standalone typed-EQ quotient lemma

GPT-5.5’s request reads (§4.5 of the third review):

The standalone lemma should list all assumptions in one place [...] and prove that quotienting preserves JEPD, strong equality, RCC5 composition, and truth of all closure formulas.

We give the lemma, listing the six load-bearing assumptions, and prove the four preservation properties.

4.1 Setup

Fix a valid finite generated split-forest certificate Q and its unfolding $\mathcal{U}(Q)$. Let Ω be the set of occurrences in $\mathcal{U}(Q)$. Each occurrence u has:

- a profile state $\text{state}(u) \in S$;
- a closure type $\text{tp}(u) := \lambda(\text{state}(u)) \in \text{Typ}(C_0)$;
- a port $\text{Port}(u) \in \text{Port}(\text{state}(u))$;
- RCC5 pair labels $\rho(u, v) \in \mathbf{B}$ for every other occurrence v , determined by the certificate's contexts, mosaics, and equality-aware reachability.

Let $\equiv_{\text{EQ}}^Q$ be the explicit port equivalence on $\bigsqcup_{s \in S} \text{Port}(s)$. Lift $\equiv_{\text{EQ}}^Q$ to occurrences by $u \equiv_{\text{EQ}} v \iff \text{Port}(u) \equiv_{\text{EQ}}^Q \text{Port}(v)$ (in the same explicit context, or transitively across explicit links).

4.2 Six load-bearing assumptions

Assumption 4.1 (Equivalence). $\equiv_{\text{EQ}}$ is an equivalence relation on Ω .

Assumption 4.2 (Vertical separation). For all $u, v \in \Omega$, if u and v are connected by strict PP- or PPI-reachability in $\mathcal{U}(Q)$ (one or more iterations of generated parent/child steps), then $u \not\equiv_{\text{EQ}} v$.

Assumption 4.3 (Type rigidity). For all $u, v \in \Omega$, if $u \equiv_{\text{EQ}} v$ then $\text{tp}(u) = \text{tp}(v)$.

Assumption 4.4 (Joint witness obligations). For all $u, v \in \Omega$ with $u \equiv_{\text{EQ}} v$, and every $\exists R.D \in \text{tp}(u) = \text{tp}(v)$ with $R \in \{\text{PP}, \text{PPI}\}$, there exists $w \in \Omega$ such that $\{u, v\} \text{Reach}_R w$ (equality-aware reachability) and $D \in \text{tp}(w)$. Equivalently: a single witness can discharge the obligation simultaneously at every $\equiv_{\text{EQ}}$ -mate of the source.

Assumption 4.5 (Quotient-compatible pair labels). For all $u, v, w \in \Omega$ with $u \equiv_{\text{EQ}} v$, $\rho(u, w) = \rho(v, w)$ and $\rho(w, u) = \rho(w, v)$.

Assumption 4.6 (Sibling-mosaic agreement on $\equiv_{\text{EQ}}$ -endpoints). For every mosaic $M \in \text{Mos}_Q$ and every pair of positions $p_1, p_2 \in P_M$ such that some realised pair of occurrences (u_1, u_2) with u_i at position p_i has $u_1 \equiv_{\text{EQ}} u_2$, the mosaic labels agree: $L_M(p_1, q) = L_M(p_2, q)$ for every $q \in P_M$, and $\tau_M(p_1) = \tau_M(p_2)$.

Where each assumption comes from in the manuscript. Assumption 4.1 is (V9.a)+(V9.d): $\equiv_{\text{EQ}}^Q$ is the reflexive-symmetric-transitive closure of explicit port links. Assumption 4.2 is (V9.b). Assumption 4.3 is (V9.a). Assumption 4.4 is the definition of Reach_R in Definition 4.5 of [1]: $s \text{Reach}_R t \iff \exists \hat{s} \in \text{Mate}_Q(s) \hat{s} \rightsquigarrow_R t$, which is exactly the joint discharge condition once lifted to occurrences. Assumption 4.5 is (V9.c). Assumption 4.6 follows from (V9.c) restricted to mosaic positions, plus support closure (SC3) which keeps mosaic overlaps consistent on $\equiv_{\text{EQ}}$ -endpoints.

4.3 The quotient

Define the quotient frame $\mathcal{U}(Q)/\equiv_{\text{EQ}}$ on equivalence classes $[u]$, with relation $\bar{\rho}([u], [v]) := \rho(u, v)$ (well-defined by Assumption 4.5) and type $\bar{\text{tp}}([u]) := \text{tp}(u)$ (well-defined by Assumption 4.3).

4.4 Four preservation properties

Lemma 4.7 (JEPD). *On $\mathcal{U}(Q)/\equiv_{\text{EQ}}$, $\bar{\rho}([u], [v])$ is a unique base relation. For $[u] \neq [v]$, $\bar{\rho}([u], [v]) \in \mathcal{B} \setminus \{\text{EQ}\}$. For $[u] = [v]$, $\bar{\rho}([u], [v]) = \text{EQ}$.*

Proof. Uniqueness. $\bar{\rho}$ is a function on $\Omega/\equiv_{\text{EQ}} \times \Omega/\equiv_{\text{EQ}}$ by Assumption 4.5.

EQ iff equal class. ($\Leftarrow$) If $[u] = [v]$, take $v' \in [u]$ and apply (M1)/reflexivity: $\rho(u, u) = \text{EQ}$, so $\bar{\rho}([u], [v]) = \rho(u, u) = \text{EQ}$. ($\Rightarrow$) Suppose $\bar{\rho}([u], [v]) = \text{EQ}$, i.e. $\rho(u, v) = \text{EQ}$. The certificate's frame axioms (F1)–(F3) give $u \equiv_{\text{EQ}} v$ when $\rho(u, v) = \text{EQ}$ under strong-EQ semantics: EQ is interpreted as identity in the model class of [1, §3.2]. Hence $[u] = [v]$. $\square$

Lemma 4.8 (Strong equality). *$\bar{\rho}([u], [v]) = \text{EQ}$ iff $[u] = [v]$, and the quotient frame is a strong abstract RCC5 frame.*

Proof. The biconditional is Lemma 4.7. For the frame property, verify (F1)–(F3) of [1, Def. 3.2]:

- (F1) $\bar{\rho}([u], [u]) = \rho(u, u) = \text{EQ}$: (M1).
- (F2) $\bar{\rho}([v], [u]) = \rho(v, u) = \text{inv}(\rho(u, v)) = \text{inv}(\bar{\rho}([u], [v]))$: (M2).
- (F3) $\bar{\rho}([u], [w]) \in \text{comp}(\bar{\rho}([u], [v]), \bar{\rho}([v], [w]))$: (M3) on $\mathcal{U}(Q)$, which holds by Theorem 3.8, projects to $\bar{\rho}$ via Assumption 4.5.

$\square$

Lemma 4.9 (RCC5 composition). *For all $[u], [v], [w] \in \mathcal{U}(Q)/\equiv_{\text{EQ}}$, $\bar{\rho}([u], [w]) \in \text{comp}(\bar{\rho}([u], [v]), \bar{\rho}([v], [w]))$.*

Proof. $\bar{\rho}([u], [w]) = \rho(u, w) \in \text{comp}(\rho(u, v), \rho(v, w)) = \text{comp}(\bar{\rho}([u], [v]), \bar{\rho}([v], [w]))$ by Theorem 3.8 and Assumption 4.5. $\square$

Lemma 4.10 (Closure-formula truth). *For every $D \in \text{Cl}(C_0)$ and every $u \in \Omega$,*

$$D \in \text{tp}(u) \iff \mathcal{U}(Q)/\equiv_{\text{EQ}}, [u] \models D.$$

Proof. By induction on $D \in \text{Cl}(C_0)$.

Atomic and negated atomic. Atomic concepts A and $\neg A$ hold at $[u]$ iff $A \in \text{tp}(u)$ or $\neg A \in \text{tp}(u)$ respectively, by definition of type and Assumption 4.3.

Boolean cases. Standard Hintikka completeness: $D_1 \sqcap D_2 \in \text{tp}(u)$ iff both $D_i \in \text{tp}(u)$; same for $\sqcup$ by Hintikka definition.

Existential $\exists R.D$. If $\exists R.D \in \text{tp}(u)$:

- $R = \text{EQ}$: (V1) normalises $\exists \text{EQ}.D \in \text{tp}(u)$ iff $D \in \text{tp}(u)$, so $[u] \models D$ by IH, hence $[u] \models \exists \text{EQ}.D$ (EQ is identity).
- $R \in \{\text{PP}, \text{PPI}\}$: (V4) plus Assumption 4.4 gives $t \in S$ with $\text{state}(u) \text{ Reach}_R t$ and $D \in \lambda(t)$. The unfolding has an occurrence w with $\text{state}(w) = t$ reachable from some $\hat{u} \equiv_{\text{EQ}} u$ by $\rightsquigarrow_R$. By Assumption 4.5 $\bar{\rho}([u], [w]) = R$, and by IH $[w] \models D$, so $[u] \models \exists R.D$.
- $R \in \{\text{DR}, \text{PO}\}$: (V6) names a mosaic position q with $L(p, q) = R$ and $D \in \tau(q)$. The unfolding realises this mosaic, giving an occurrence w at q with $\rho(u, w) = R$ and $\text{tp}(w) \ni D$. By IH $[w] \models D$, so $[u] \models \exists R.D$.

Conversely, if $[u] \models \exists R.D$, there is $[w]$ with $\bar{\rho}([u], [w]) = R$ and $[w] \models D$. By IH $D \in \text{tp}(w)$. If $\exists R.D \notin \text{tp}(u)$, then since $\text{Cl}(C_0)$ is closed under NNF complement, $\forall R.\bar{D} \in \text{tp}(u)$. By (V5)/(V7) (universal safety), $\bar{D} \in \text{tp}(w)$, contradicting $D \in \text{tp}(w)$ and the Hintikka property.

Universal $\forall R.D$. Dual: $\forall R.D \in \text{tp}(u)$ forces D into every R -target's type by (V5)/(V7), so every R -successor $[w]$ of $[u]$ satisfies D by IH. Conversely, if $\forall R.D \notin \text{tp}(u)$ then $\exists R.\bar{D} \in \text{tp}(u)$, and the existential case gives some $[w]$ with $\bar{D} \in \text{tp}(w)$, so $[w] \not\models D$, so $[u] \not\models \forall R.D$. $\square$

Theorem 4.11 (Typed-EQ quotient lemma). *Let Q be a valid finite generated split-forest certificate and $\equiv_{\text{EQ}}$ the lift of $\equiv_{\text{EQ}}^Q$ to occurrences. Under Assumptions 4.1–4.6, the quotient $\mathcal{U}(Q)/\equiv_{\text{EQ}}$ is a strong abstract RCC5 frame, and for every $D \in \text{Cl}(C_0)$ and $u \in \Omega$, $D \in \text{tp}(u)$ iff $\mathcal{U}(Q)/\equiv_{\text{EQ}}, [u] \models D$. In particular, $\mathcal{U}(Q)/\equiv_{\text{EQ}}, [u_0] \models C_0$, where u_0 is the distinguished root occurrence.*

Proof. Lemmas 4.7–4.10. $\square$

Remark 4.12 (Discharging C4). *Theorem 4.11 closes the obligation C4 Typed-EQ quotient soundness. The verification status of C4 moves from **open** to **closed (paper)**. Lean target L4 in the verification recommendations remains open.*

5 Item 6: finite certificate grammar and enumeration bound

GPT-5.5's request reads (§4.6 of the third review):

The manuscript should include a compact formal grammar for certificates and a computable bound on local closure types, equality-mate interface ports, selected-parent and selected-child context slots, request summaries, sibling/support mosaics and their maximum arity, and finite graph size after blocking. The bound may be coarse but should be explicit enough that enumeration is a mathematical consequence.

We give the grammar and derive the bound.

5.1 Grammar of certificate components

Fix the input C_0 and set $n := |\text{Cl}(C_0)| \leq 2|\text{sub}(C_0)|$. A finite generated split-forest certificate is the formal expression of the following abstract grammar (writing $\sqcup$ for choice and $\langle \cdot, \dots, \cdot \rangle$ for tuples):

Certificate $Q ::= \langle S, s_0, \lambda, \text{par}_Q, \text{Ch}_Q, \text{Ctx}_Q, \text{Mos}_Q, E_Q, \text{wit}_Q \rangle$
 StateSet $S ::= \emptyset \sqcup \{\sigma\} \cup S, \quad \sigma \in \Sigma$
 Profile $\sigma ::= \langle \tau, \kappa, \mu, \eta \rangle$
 Type $\tau \subseteq \text{Cl}(C_0), \quad \tau \in \text{Typ}(C_0)$
 ContextColour $\kappa ::= \langle \kappa_\uparrow, \kappa_\downarrow, \kappa_{\text{side}} \rangle$
 SideMenu $\mu ::= \{\exists R.D \mapsto (M_i, q_i) : \exists R.D \in \tau, R \in \{\text{DR}, \text{PO}\}\}$
 InterfaceSummary $\eta ::= \{(\pi, R) \mapsto \text{Reach}_R(\pi) \subseteq \text{Cl}(C_0) : \pi \in \text{Port}(\sigma), R \in \{\text{PP}, \text{PPI}\}\}$
 ParentMap $\text{par}_Q ::= S \rightarrow S$
 ChildMultiset $\text{Ch}_Q(s) \subseteq_{\text{mset}} S$
 Context $\text{Ctx}_Q ::= \{s \mapsto \text{context scheme}\}$
 Mosaic $M ::= \langle P, \tau_M, L_M \rangle$
 PositionSet $P \subseteq \{1, \dots, m(n)\}$
 Edge $L_M(p, q) \in \mathbb{B}$
 EqPort $E_Q ::= \{(\sigma_i, p_i) \sim (\sigma_j, p_j)\}$
 Witness $\text{wit}_Q ::= \{\exists R.D \mapsto \text{mechanism}\}$

Each terminal in the grammar ranges over a finite alphabet determined by n :

- $\text{Cl}(C_0)$: $\leq 2 \cdot |\text{sub}(C_0)| \leq 2n$.
- $\text{Typ}(C_0)$: $\leq 2^n$.
- $\text{Port}(\sigma)$: bounded by a polynomial; we set $|\text{Port}(\sigma)| \leq n + 1$ (one port per PP/PPI-mate obligation, plus the identity port).
- Context colours κ : at most three polynomial slots, each carrying one of finitely many context scheme names.
- Mos_Q : mosaics with position count $\leq m(n)$, where $m(n)$ is the manuscript's crude width (one source, one position per existential formula in τ , one equality-mate boundary position per containment witness, and one position per realised triple).

We give an explicit polynomial $p(n)$ below such that every component alphabet has cardinality $\leq 2^{p(n)}$.

5.2 Explicit component bounds

Fix a polynomial bound $m(n) = c \cdot n^k$ for the mosaic position count, with c, k determined by the manuscript's context schemes (e.g. $c = 8, k = 2$ suffices: each existential formula $\exists R.D \in \tau$ contributes one source position and one witness position, and triple closure quadruples this). Then:

- **Closure types.** $|\text{Typ}(C_0)| \leq 2^n$.
- **Equality-mate interface ports.** $|\text{Port}(\sigma)| \leq n + 1$, so the port alphabet across profiles has size $\leq |\Sigma| \cdot (n + 1)$.

- **Selected-parent and selected-child slots.** par_Q is a partial function $S \rightarrow S$, giving $|S|^{|S|+1}$ choices. Ch_Q is a multiset on S with capacity bounded by the existentials in $\lambda(s)$, so $|\text{Ch}_Q(s)| \leq n$ and the number of child assignments per state is $\leq n \cdot |S|^n$.
- **Request summaries.** η records, per port per vertical direction, a subset of $\text{Cl}(C_0)$: $(n+1) \cdot 2 \cdot 2^{2n} = O(n \cdot 2^{2n})$ bits per profile.
- **Sibling/support mosaics.** A mosaic is a function $\tau_M : P \rightarrow \text{Typ}(C_0)$ and $L_M : P^2 \rightarrow \mathbb{B}$: $\leq (2^n)^{m(n)} \cdot 5^{m(n)^2} \leq 2^{p_1(n)}$ for $p_1(n) := n \cdot m(n) + m(n)^2 \cdot \log_2 5$. Mos_Q is a subset of all such mosaics: $\leq 2^{2^{p_1(n)}}$ families. Maximum arity is $m(n)$.
- **Finite graph size after blocking.** By the finite-regularisation lemma (Lemma 7.3 of [1]) the certificate's state set S has $|S| \leq N(C_0)$, the number of complete boundary profiles:

$$|\Sigma| \leq |\text{Typ}(C_0)| \cdot (\text{context colour count}) \cdot (\text{side menu count}) \cdot (\text{interface count})$$

Each factor is bounded above by $2^{p_2(n)}$ for an explicit polynomial p_2 , so $|\Sigma| \leq 2^{p(n)}$ for some polynomial $p(n)$, and $|S| \leq |\Sigma| \leq 2^{p(n)}$.

Collecting these,

$$p(n) := n + 2 \cdot m(n) \cdot \log_2 5 + 2n \cdot (n+1) + \text{const}$$

is a polynomial of degree $\max(k+1, 2k)$ in n .

5.3 Enumeration bound

Lemma 5.1 (Certificate enumeration bound). *There is an explicit polynomial $p(n)$ such that the number of finite generated split-forest certificates for C_0 with state count $\leq B(C_0) := 2^{2^{p(n)}}$ and context width $\leq B(C_0)$ is at most $2^{2^{q(n)}}$ for some polynomial $q(n)$. Each certificate's validity (V1)–(V10) is decidable in time polynomial in its description size.*

Proof. By the grammar above, a certificate is a finite tuple over alphabets of sizes $\leq 2^{p(n)}$ (states, types, contexts, mosaic positions, port symbols, witness mechanisms) with at most $B(C_0) = 2^{2^{p(n)}}$ components. The total number of certificates of size $\leq B(C_0)$ is then bounded by

$$(2^{p(n)})^{B(C_0)} \cdot (\text{combinatorial overhead}) \leq 2^{p(n) \cdot 2^{2^{p(n)}}} \leq 2^{2^{q(n)}}$$

for some polynomial q .

Each validity clause (V1)–(V10) is a finite-quantifier predicate over the certificate's components: (V1)–(V3) are local type/context checks, (V4)–(V5) are graph-reachability checks on S (linear in $|S|$ per query), (V6)–(V8) are mosaic-position membership and triple checks (polynomial in mosaic count), (V9) is a congruence check on the explicit-port relation, (V10) is exact summary computation by reachability. All in time polynomial in $|Q|$, which is polynomial in $B(C_0)$. $\square$

Theorem 5.2 (Effective decidability). *Concept satisfiability for $\mathcal{ALCT}_{\text{RCC5}}$ is decidable by enumerating finite generated split-forest certificates up to size $B(C_0) = 2^{2^{p(n)}}$ and checking validity. Soundness and completeness are provided by Theorems 6.1 and 7.4 of [1]; the patchwork and quotient steps used inside those theorems are discharged by Theorems 3.8 and 4.11 of the present addendum.*

Proof. Lemma 5.1 enumerates certificates; Theorem 3.8 and Theorem 4.11 discharge the only steps in the manuscript’s proof of Theorems 6.1 and 7.4 of [1] that the third-round review left open; Lemma 7.3 (finite regularisation) of [1] provides the bound $|S| \leq B(C_0)$ on representative certificates. $\square$

Remark 5.3 (Discharging C8). *With the grammar above and Lemma 5.1, the obligation C8 Effective finite certificate language moves from **open** to **closed (paper)**. Lean target L3 (enumeration bound mechanization) remains open.*

6 Status after the addendum

ID	Obligation	Status after ad- dendum	Source
C1	RCC5 composition table	PASS	WP1 + [1] §3.1
C2	Closure and witness thinning	partial	Hand proof in [1] Lem. 3.3; Lean- pending
C3	Blocking separated from semantic EQ	partial to PASS	(V9) of [1] Def. 4.5; WP2 corpus
C4	Typed-EQ quotient soundness	closed (paper) (paper)	Theorem 4.11 of this addendum
C5	Multiple upward PP-witnesses	PASS	v2 §3.2 (C_{AB}^{inc})
C6	Request-closed cycles	PASS	v2 §3.3 (pure-isolation test)
C7	Support-closed mosaic patchwork	closed (paper) (paper)	Theorem 3.8 of this addendum + WP6
C8	Effective finite certificate language	closed (paper) (paper)	Lemma 5.1 of this addendum

What remains open. At paper level only C2 is still **partial**, and only because the witness-thinning lemma of [1] Lem. 3.3 is hand-proved and would benefit from a fully expanded proof. All other obligations are either **PASS** (computational) or **closed (paper)** (paper). The Lean targets L1–L7 from the verification recommendations remain open, in the priority order L1, L3, L4, L6, then L2/L5/L7.

Honest assessment. The addendum upgrades C4/C7/C8 from *computationally supported, theorem pending* to *computationally supported, paper theorem closed, Lean pending*. It does not claim Lean-level certainty, and it does not amplify the manuscript’s complexity bound: $B(C_0) = 2^{2^{p(n)}}$ is unchanged. The addendum’s contribution is making each load-bearing step explicit so that a Lean formalisation has a definite target.

References

- [1] *A Generated Split-Forest Decidability Proof for $\mathcal{ALCI}_{\text{RCC5}}$ Without Automata: A Repaired Containment-Oriented Presentation.* GPT-5.5, May 2026. In repository at `papers/gpt5.5_round2/repaired_split_forest_no_automata_proof.tex`.

- [2] Jochen Renz and Bernhard Nebel. *On the Complexity of Qualitative Spatial Reasoning: A Maximal Tractable Fragment of the Region Connection Calculus*. Artificial Intelligence 108 (1–2), 1999, pp. 69–123.
- [3] Michael Wessel. *Qualitative Spatial Reasoning with the $ALCI_{RCC}$ Family of Description Logics*. Technical report, University of Hamburg, 2002/2003.